

11.17 The equations (11.68) and (11.69) in the book are incorrect. The correct ones as per official errata are:

$$(11.68) = \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}') + H(\mathcal{R}))\}$$

$$(11.69) = \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}) + H(\mathcal{R}'))\}$$

We consider three cases.

1. $H(\mathcal{R}') = H(\mathcal{R})$

$$\begin{aligned} (11.68) &= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \min\{1, \exp(0)\} \\ &= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \end{aligned}$$

$$\begin{aligned} (11.69) &= \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \min\{1, \exp(0)\} \\ &= \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \\ &= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \end{aligned}$$

Thus, (11.68) = (11.69).

2. $H(\mathcal{R}') > H(\mathcal{R})$

$$\begin{aligned} (11.68) &= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}') + H(\mathcal{R}))\} \\ &= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \exp(-H(\mathcal{R}') + H(\mathcal{R})) \\ &= \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \end{aligned}$$

$$(11.69) = \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}) + H(\mathcal{R}'))\}$$

$$= \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2}$$

Thus, (11.68) = (11.69).

3. $H(\mathcal{R}) > H(\mathcal{R}')$

$$(11.68) = \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}') + H(\mathcal{R}))\}$$

$$= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2}$$

$$(11.69) = \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \min\{1, \exp(-H(\mathcal{R}) + H(\mathcal{R}'))\}$$

$$= \frac{1}{Z_H} \exp(-H(\mathcal{R}')) \delta V \frac{1}{2} \exp(-H(\mathcal{R}) + H(\mathcal{R}'))$$

$$= \frac{1}{Z_H} \exp(-H(\mathcal{R})) \delta V \frac{1}{2}$$

Thus, (11.68) = (11.69).